

KHANBOGD, Mongolia

WMO: 443850

Lat: 43.1957N

Lon: 107.1940E

Elev: 914

StdP: 90.82

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.5	(c) -20.5	(d) -30.4	(e) 0.3	(f) -18.1	(g) -28.2	(h) 0.3	(i) -16.3	(j) 14.2	(k) -9.5	(l) 12.4	(m) -7.9	(n) 2.8	(o) 270	(p) 0.604

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.2	(c) 34.6	(d) 17.4	(e) 32.9	(f) 16.8	(g) 31.4	(h) 16.2	(i) 19.6	(j) 30.1	(k) 18.5	(l) 29.2	(m) 17.7	(n) 28.1	(o) 4.4	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 16.4	(b) 13.0	(c) 23.0	(d) 14.9	(e) 11.8	(f) 23.3	(g) 13.5	(h) 10.8	(i) 23.3	(j) 60.5	(k) 30.7	(l) 56.4	(m) 29.0	(n) 53.6	(o) 28.3	(p) 24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.9	(b) 12.0	(c) 10.2	(d) DB	(e) -25.2	(f) 37.5	(g) 2.2	(h) 1.9	(i) -26.8	(j) 38.9	(k) -28.1	(l) 40.0	(m) -29.3	(n) 41.1	(o) -30.9	(p) 42.5	(q) 25.5
			(d) WB	(e) -25.5	(f) 21.2	(g) 2.1	(h) 1.7	(i) -27.0	(j) 22.4	(k) -28.3	(l) 23.4	(m) -29.4	(n) 24.3	(o) -31.0	(p) 25.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	-11.1	-7.5	1.4	10.1	16.8	22.5	25.5	23.5	17.5	8.7	-1.4	-9.4
	DBStd	13.76	5.18	6.00	6.27	5.78	5.16	3.93	3.22	3.41	4.53	5.45	5.67	4.79
	HDD10.0	2533	654	490	273	68	8	0	0	0	5	88	344	603
	HDD18.3	4337	912	723	525	250	91	11	1	3	68	299	593	861
	CDD10.0	1847	0	0	7	72	218	374	480	418	229	48	1	0
	CDD18.3	610	0	0	0	4	43	134	222	162	42	2	0	0
	CDH23.3	6229	0	0	0	47	467	1391	2456	1566	293	11	0	0
	CDH26.7	2435	0	0	0	8	143	534	1101	597	51	1	0	0
(14) Wind	WSAvg	4.4	4.2	4.2	4.8	5.2	5.4	4.2	3.8	3.5	3.8	4.2	4.7	4.5
(15) Precipitation	PrecAvg	109	2	1	3	4	11	20	27	24	10	4	2	1
	PrecMax	175	4	7	20	16	74	57	77	80	51	16	9	7
	PrecMin	71	0	0	0	0	0	1	1	0	1	0	0	0
	PrecStd	29	1	2	5	5	17	14	17	20	10	4	2	2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	9.1	19.5	27.9	33.0	35.6	37.5	35.6	29.9	24.7	14.2	3.8
		MCWB	-1.4	1.9	6.8	11.6	13.8	16.7	19.0	18.0	14.2	11.0	5.2	-2.0
	2%	DB	0.7	6.3	16.1	24.4	30.2	33.3	35.2	33.4	28.1	21.7	11.2	0.9
		MCWB	-3.7	0.0	5.1	9.7	12.7	15.9	17.8	17.7	13.8	9.7	3.5	-3.4
	5%	DB	-2.0	3.8	13.6	22.1	27.9	31.5	33.3	31.7	26.6	19.1	8.7	-1.0
		MCWB	-5.6	-1.6	3.9	8.8	11.9	15.2	17.5	16.6	13.3	8.3	1.9	-4.8
	10%	DB	-4.0	1.4	10.9	19.6	25.9	29.7	31.9	29.9	25.0	16.9	6.6	-2.8
		MCWB	-7.0	-3.4	2.4	7.4	11.0	14.6	17.0	16.2	12.6	7.1	0.6	-6.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.3	2.4	7.1	12.0	14.7	18.7	21.3	21.4	17.1	12.1	5.6	-1.6
		MCDB	3.5	7.8	18.2	26.0	29.3	29.2	32.9	27.3	24.6	22.0	13.9	3.3
	2%	WB	-3.5	0.1	5.4	10.2	13.5	17.5	20.0	19.6	15.6	10.4	3.7	-3.3
		MCDB	0.4	6.2	15.7	22.7	27.8	29.1	30.7	29.3	24.9	20.8	11.0	0.7
	5%	WB	-5.5	-1.5	4.0	9.0	12.5	16.6	18.8	18.5	14.5	8.7	2.2	-4.7
		MCDB	-2.0	3.8	13.3	21.7	26.4	28.1	30.1	28.6	24.2	18.3	8.7	-1.1
	10%	WB	-7.0	-3.3	2.5	7.7	11.5	15.8	18.0	17.4	13.4	7.5	0.7	-6.0
		MCDB	-4.1	1.3	10.7	19.2	24.7	27.3	28.6	27.4	23.0	16.4	6.7	-2.8
(35) Mean Daily Temperature Range	MDBR		8.7	10.5	11.5	11.5	11.7	10.7	10.2	10.2	10.2	10.2	8.9	7.9
	5% DB	MCDBR	10.0	12.3	13.2	13.6	13.1	11.9	11.1	11.3	11.1	11.5	10.4	9.0
		MCWBR	7.4	8.6	7.8	7.4	6.3	5.1	4.2	4.6	5.1	6.4	6.5	6.6
	5% WB	MCDBR	10.0	11.9	13.1	13.1	12.7	11.0	10.2	10.0	10.1	11.5	10.3	9.0
(40) Clear-Sky Solar Irradiance	taub		0.283	0.315	0.360	0.411	0.447	0.471	0.447	0.426	0.386	0.340	0.299	0.281
	taud		2.275	2.201	2.095	2.004	1.959	1.984	2.120	2.154	2.202	2.262	2.297	2.270
	Ebn at Noon		845	871	870	851	829	808	824	828	835	831	816	808
	Edh at Noon		86	110	140	167	180	176	152	142	125	103	83	79
(44) All-Sky Solar Radiation	RadAvg		2.35	3.45	4.96	6.19	7.02	6.86	6.55	6.08	5.14	3.94	2.64	2.03
	RadStd		0.19	0.21	0.30	0.25	0.32	0.32	0.30	0.24	0.20	0.12	0.15	0.17

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page